MERAKI - BIOS : RE-NEET 2026
TARGET 360/360 IN BIOLOGY
THREE [3] ALL INDIA FREE TESTS IN BIOLOGY
27.05.2026; 07.06.2026, 17.06.2026

PRACTICE TEST: REVISE AND RECALL: CELL CYCLE AND CELL DIVISION

1. Consider the given two statements:

Statement I: Cell growth in terms of cytoplasmic increase is a continuous process, whereas DNA synthesis occurs only during a specific stage of the cell cycle.

Statement II: The cell cycle ensures coordinated cell division, DNA replication, and cell growth so that daughter cells receive intact genomes.

1. Statement I is incorrect; Statement II is incorrect
2. Statement I is incorrect; Statement II is correct
3. Statement I is correct; Statement II is incorrect
4. Statement I is correct; Statement II is correct

[Page 120 - Recall - what happens in a prokaryotic cell - do you know what are the stages?]

2. During the S phase of the cell cycle, if a cell initially has a DNA content of 2C and 2n chromosomes, what will be the DNA content and chromosome number at the END of the S phase?

1. DNA = 4C, Chromosomes = 4n
2. DNA = 2C, Chromosomes = 2n
3. DNA = 4C, Chromosomes = 2n
4. DNA = 2C, Chromosomes = 4n

[Page 121 - very important question - what happens when centromere splits and chromatids get separated?]

3. Which of the following statements correctly describes cells in the G_0 (quiescent) stage?

1. Cells in G_0 are metabolically inactive and have permanently lost the ability to divide under any circumstance
2. Cells in G_0 are metabolically active but do not proliferate unless required by the organism
3. Cells enter G_0 directly after completing DNA replication in S phase
4. G_0 cells actively divide to replace damaged or lost cells on a continuous basis

[Page 122 - very frequently asked - recall does the cell exit the cell cycle or suspends the cell cycle when it enters G_0]

4. Which of the following sets of cell organelles/structures disappear by the end of prophase during mitosis?

1. Nucleolus, Nuclear envelope, Golgi complex, Endoplasmic reticulum
2. Golgi complex, Endoplasmic reticulum, Nucleolus, Cell membrane
3. Mitochondria, Nucleolus, Nuclear envelope, Ribosomes
4. Nuclear envelope, Centrioles, Golgi complex, Mitochondria

[Page 123; Why is Nuclear Envelope BreakDown essential]

5. Consider the given two statements:

Statement I: Metaphase plate is the plane of alignment of chromosomes at the equator, where each kinetochore is attached to spindle fibres from both poles simultaneously.

Statement II: During anaphase, the number of chromosomes in the cell effectively doubles as each chromatid is now considered an independent daughter chromosome moving toward opposite poles.

1. Statement I is incorrect; Statement II is incorrect
2. Statement I is incorrect; Statement II is correct
3. Statement I is correct; Statement II is incorrect
4. Statement I is correct; Statement II is correct

[Pages 123/124; Think - correlate with question 2]

6. Which of the following correctly distinguishes cytokinesis in plant cells from animal cells?

1. Animal cells form a cell plate at the centre while plant cells form a cleavage furrow inward
2. Plant cells form a cell plate growing from centre outward, while animal cells form a furrow in the plasma membrane that deepens inward
3. Both plant and animal cells undergo cytokinesis by forming a cleavage furrow due to flexible cell walls
4. Plant cells divide by forming a furrow while animal cells form a cell plate due to the presence of a rigid cell wall

7. Chiasmata visible during diplotene represent:

1. Sites where synaptonemal complex is formed between homologous chromosomes
2. Points where homologous chromosomes remain attached after crossing over, appearing as X-shaped structures
3. Regions where sister chromatids separate from each other during recombination
4. Sites of kinetochore attachment to spindle fibres during diplotene

8. Which of the following correctly describes the key events of meiosis?

1. Meiosis involves a single cycle of DNA replication and two sequential cycles of cell division, ultimately producing four haploid cells

2. Meiosis involves two cycles of DNA replication and two cycles of cell division, producing four haploid daughter cells

3. Meiosis involves a single cycle of DNA replication and a single cycle of cell division, producing two haploid daughter cells

4. Meiosis involves two cycles of DNA replication but only one cycle of cell division, restoring the diploid number at the end

[Page 125 - Recall - What is the critical difference between interkinesis and interphase]

9. Which of the following correctly describes the events following Telophase I in meiosis?

1. DNA replication occurs during interkinesis to prepare the cell for Meiosis II

2. Nuclear membrane disappears at Telophase I and does not reappear until the completion of Meiosis II

3. Interkinesis is a prolonged stage where chromosomes fully extend to the interphase state before Prophase II begins

4. Cytokinesis follows Telophase I forming a dyad of cells, after which interkinesis occurs without any DNA replication before Prophase II

10. Consider the given two statements:

Statement I: Meiosis conserves the chromosome number of a species across generations in sexually reproducing organisms.

Statement II: Meiosis increases genetic variability in a population from one generation to the next.

Which of the following is correct?

1. Both Statement I and Statement II are true, and Statement II is the correct explanation of Statement I.

2. Both Statement I and Statement II are true, but Statement II is not the correct explanation of Statement I.

3. Statement I is true, but Statement II is false.

4. Statement I is false, but Statement II is true.

[Page 128 - meiosis halves the chromosome number during gamete formation, and upon fertilisation the diploid number is restored, thus conserving the species-specific chromosome number across generations]